

8

a

i

A species of nucleus with a specific value of proton and neutron number.

ii

Number of nuclei disintegrating per unit time / rate of nuclear disintegration

iii

Nucleus' probability of decay per unit time.

b

i

9.0	98	1.99
10.0	68	1.83
12.0	33	1.52

Correct values

Correct precision

ii

1. Point correctly plotted with precision of half the smallest square (2, 3.13)

2. Line of best fit.

iii

The substance contains a mixture of elements with different decay constants.

iv

Correct gradient calculations with coordinates read correctly

Correct units h⁻¹

v

It represents the decay constant of one of the elements within the mixture multiplied by lg e

c

i

To kill the bacteria on the food thus allowing them to last longer.

ii

${}^{60}_{27}\text{Co} \rightarrow {}^{60}_{28}\text{Ni} + {}^0_{-1}\text{e}^- (+ \bar{\nu}_e + \gamma)$ optional

iii

$A = A_0 e^{-\lambda t}$
 $A = 50000 e^{-\frac{\ln 2}{5.271} 5}$
 $= 25900 \text{ Ci (25906.99)}$

Since the activity now is lesser than 35000, it is too weak a source and unusable

iv

$$A = \lambda N = \frac{\ln 2}{t_{1/2}} N$$
$$N = \frac{50000 \times 3.7 \times 10^{10}}{\frac{\ln 2}{5.271 \times 365 \times 24 \times 60 \times 60}} = 4.4366 \times 10^{23}$$

Total mass = N * mass of 1 cobalt atom = N * 60 u = 4.4366 × 10²³ × 1.66 × 10²⁷ * 1000

Total mass = 44.2 g